

05461

MASTER OF ARTS (PSYCHOLOGY)

Term-End Examination

December, 2016

MPC-006 : STATISTICS IN PSYCHOLOGY

Time : 2 hours

Maximum Marks : 50

Note : (i) *All sections are compulsory.*

(ii) *Use of simple calculator be permitted.*

SECTION - A

Note : Answer **any two** of the following questions
in about **500** words each : **2x10=20**

1. Discuss the concept of Normal Curve. Describe 3+7
properties of Normal Probability Curve.
2. Define Non-parametric Statistics. Describe the 3+7
assumptions and use of non-parametric tests.
3. A research was carried out to find if significant 10
difference exists in the self concept of early,
middle and late adolescents. The scores obtained
on self concept are given below. Using ANOVA
indicate if the groups differ on self concept
significantly.

Group I (Early)	Group II (Middle)	Group III (Late)
14	8	7
15	13	5
13	14	7
12	22	6
11	14	8
10	24	8
9	12	10
5	15	8
3	20	6
4	15	6

Critical value : 5.49 at 0.01 level of significance.

3.35 at 0.05 level of significance.

4. Compute regression equation for X and Y based on the data given below : 10

Individuals	X	Y
A	2	10
B	7	12
C	8	3
D	3	10
E	5	10

SECTION - B

Note : Answer **any four** of the following questions
in about **300** words each : **4x6=24**

5. Define hypothesis testing. Discuss general 2+4
procedure for testing a hypothesis with the help
of suitable example.

6. Calculate Mann-Whitney U-test with the help of 6
the following data :

Group 1 : 40, 17, 46, 51, 45

Group 2 : 12, 18, 20, 15, 17

7. Compute Chi-square for the following data : 6

Gender	Answers given	
	Correct	Incorrect
Males	50	60
Females	40	30

8. Compute Spearman's Rank Correlation for the 6
following data :

Data 1 : 44, 45, 45, 34, 43, 23, 54, 34, 67, 45

Data 2 : 12, 21, 32, 12, 12, 15, 26, 12, 16, 12

9. Describe point biserial correlation and tetrachoric 3+3
correlation.

SECTION - C

Note : Write short notes on **any two** of the following in about **100** words each : **2x3=6**

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|-----|---|---|
| 10. | Kruskal-Walli's ANOVA Test | 3 |
| 11. | Levels of measurement | 3 |
| 12. | Wilcoxon Matched Pair Signed Ranks Test | 3 |
